

# Exceptions

15-3-26

Aim – Demonstrate exceptions in python.

Theory – Exception handling in Python is used to manage runtime errors without stopping program execution. The try block contains code that may cause errors, while the except block handles specific exceptions like ZeroDivisionError, ValueError, IndexError, KeyError, and FileNotFoundError. Custom exceptions allow programmers to define and handle application-specific errors.

A1.

# 1. ZeroDivisionError

try:

```
    result = 10 / 0
```

except ZeroDivisionError as e:

```
    print("ZeroDivisionError:", e)
```

# 2. ValueError

try:

```
    num = int("abc")
```

except ValueError as e:

```
    print("ValueError:", e)
```

# 3. IndexError

try:

```
    lst = [1, 2, 3]
```

```
    print(lst[5])
```

except IndexError as e:

```
    print("IndexError:", e)
```

# 4. KeyError

try:

```
    d = {'a': 1}
```

```
    print(d['b'])
```

except KeyError as e:

```
    print("KeyError:", e)
```

## # 5. FileNotFoundError

try:

```
with open("nonexistentfile.txt", "r") as f:  
    content = f.read()
```

except FileNotFoundError as e:

```
    print("FileNotFoundError:", e)
```

## # Custom Exception

```
class MyCustomException(Exception):
```

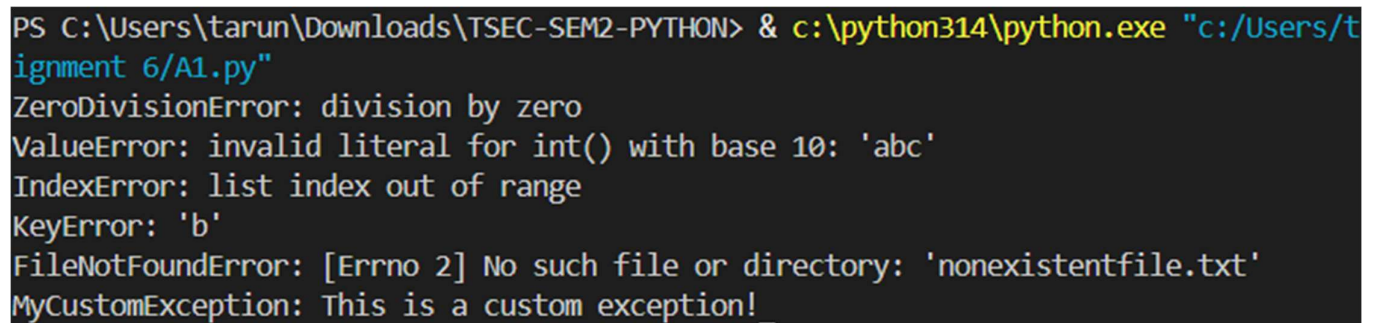
```
    pass
```

try:

```
    raise MyCustomException("This is a custom exception!")
```

except MyCustomException as e:

```
    print("MyCustomException:", e)
```



```
PS C:\Users\tarun\Downloads\TSEC-SEM2-PYTHON> & c:\python314\python.exe "c:/Users/tarun/Downloads/TSEC-SEM2-PYTHON/Assignment 6/A1.py"  
ZeroDivisionError: division by zero  
ValueError: invalid literal for int() with base 10: 'abc'  
IndexError: list index out of range  
KeyError: 'b'  
FileNotFoundError: [Errno 2] No such file or directory: 'nonexistentfile.txt'  
MyCustomException: This is a custom exception!
```

## Conclusion

Exception handling improves program reliability by preventing crashes and allowing developers to manage different runtime errors in a controlled way.